so that $E_h^\alpha(R_h' y_h) \to E_i^{CVK}(v')$. Moreover,

$$\frac{1}{h^\alpha}\int_\Omega f_h \cdot R_h' y_h' - \frac{1}{h^\alpha} F_h(R_h' U_i) = \frac{1}{h^\alpha}\int_S f_h \cdot R_h' U_i^{-1} \begin{pmatrix} h^{2\gamma-2} u_h' \\ h^{\gamma-1} v_h' \end{pmatrix} dx$$
$$\to \int_S f \cdot R' U_i^{-1} e_3 v' \, dx \,,$$

where u_h', v_h' are defined as in (4)–(5) with y_h' and U_i in place of y_h and U_j, respectively. By hypothesis (F1), we have that $R_h' U_i \in \mathcal{M}_h$ for $h \ll 1$. Thus,

$$F_h(P_h^j(\bar{R}_h^T) U_j) = F_h(R_h' U_i) \,,$$

and $F(R'(W')^2 U_i) \le 0$. Hence, by (44) we deduce that

$$\begin{aligned} J_j^{CVK}(v, \bar{R}^T, \beta W) &\le \liminf_{h\to 0} \left(J_h^\alpha(y_h) + \frac{1}{h^\alpha} F_h(P_h^j(\bar{R}_h^T) U_j) \right) \\ &\le \limsup_{h\to 0} \left(J_h^\alpha(y_h) + \frac{1}{h^\alpha} F_h(P_h^j(\bar{R}_h^T) U_j) \right) \\ &= \limsup_{h\to 0} \left(\inf_y J_h^\alpha(y) + \frac{1}{h^\alpha} F_h(P_h^j(\bar{R}_h^T) U_j) \right) \\ &\le \limsup_{h\to 0} \left(J_h^\alpha(R_h' y_h') + \frac{1}{h^\alpha} F_h(R_h' U_i) \right) \\ &= J_i^{CVK}(v', R', 0) \le J_i^{CVK}(v, R', W') \,, \end{aligned}$$

that gives the minimality property. ■

A Construction of isometries

In this appendix we collect a few results regarding the construction of isometries, mainly based on refinements of [15, Section 8].

Lemma A.1. *Let $U \in \mathbb{R}^{3\times 3}$ be a symmetric and positive definite matrix. For $\varepsilon > 0$ let $y_\varepsilon \in W^{2,2}(S;\mathbb{R}^3)$ be a map of the form*

$$y_\varepsilon(x') = U \begin{pmatrix} x' \\ 0 \end{pmatrix} + \varepsilon v U^{-1} e_3 + \varepsilon^2 U \begin{pmatrix} u_\varepsilon \\ 0 \end{pmatrix} \,,$$

such that $\nabla y^T \nabla y = (U^2)'$, where $v \in W^{2,\infty}(S)$ and u_ε is a bounded sequence in $W^{2,\infty}(S;\mathbb{R}^2)$. Then, if ν_ε is defined as in Lemma 3.6, we have

$$\nabla y_\varepsilon^T \nabla \nu_\varepsilon = -\varepsilon \nabla^2 v + O(\varepsilon^2) \,.$$

Proof. First, we compute $\partial_1 y_\varepsilon \wedge \partial_2 y_\varepsilon$. We have

$$\partial_1 y_\varepsilon \wedge \partial_2 y_\varepsilon = U e_1 \wedge U e_2 + \varepsilon \left[U e_1 \wedge U^{-1} \begin{pmatrix} 0 \\ \partial_2 v \end{pmatrix} + U^{-1} \begin{pmatrix} 0 \\ \partial_1 v \end{pmatrix} \wedge U e_2 \right] + O(\varepsilon^2) \,.$$

Given the identity $Ua \wedge Ub = \det(U) U^{-1}(a \wedge b)$, we easily deduce that

$$\begin{aligned} \partial_1 y_\varepsilon \wedge \partial_2 y_\varepsilon &= U e_1 \wedge U e_2 \\ &\quad + \varepsilon \det(U) U^{-1} \left[\partial_2 v e_1 \wedge (U^{-1})^2 e_3 + \partial_1 v (U^{-1})^2 e_3 \wedge e_2 \right] + O(\varepsilon^2) \,. \end{aligned}$$

Computing the cross products, we get

$$\partial_1 y_\varepsilon \wedge \partial_2 y_\varepsilon = Ue_1 \wedge Ue_2 - \varepsilon \det(U) U^{-1} \begin{pmatrix} |U^{-1}e_3|^2 \partial_1 v \\ |U^{-1}e_3|^2 \partial_2 v \\ -\sum_{k=1}^2 (U^{-1}e_k \cdot U^{-1}e_3)\partial_k v \end{pmatrix} + O(\varepsilon^2)\,.$$

Let us set $q(x') := -\frac{1}{|U^{-1}e_3|^2} \sum_{k=1}^2 (U^{-1}e_k \cdot U^{-1}e_3)\partial_k v$. We have

$$\nabla(\partial_1 y_\varepsilon \wedge \partial_2 y_\varepsilon) = -\varepsilon |U^{-1}e_3|^2 \det(U) U^{-1} \begin{pmatrix} \nabla^2 v \\ \nabla q \end{pmatrix} + O(\varepsilon^2)\,.$$

Now observe that for every triplet of indices $i, j, k = 1, 2$ we have

$$\partial_i y_\varepsilon \cdot \partial^2_{jk} y_\varepsilon = O(\varepsilon^2)\,.$$

Moreover,

$$\partial_i y_\varepsilon^T U^{-1} = e_i^T + O(\varepsilon)\,.$$

Combining the previous equations with the definition of ν_ε, the thesis follows. ■

Lemma A.2. *Let $U \in \mathbb{R}^{3\times 3}$ be a symmetric and positive definite matrix and define $A = \sqrt{(U^2)'}$. Suppose that S is simply connected. Let $v \in W^{2,2}(S)$ be such that $|U^{-1}e_3| \|\nabla v A^{-1}\|_{L^\infty} < 1$. Then, there exists $\phi \in W^{2,2}(S;\mathbb{R}^2)$ such that the map*

$$y(x') = vU^{-1}e_3 + U \begin{pmatrix} \phi \\ 0 \end{pmatrix} \tag{45}$$

satisfies $\nabla y^T \nabla y = (U^2)'$ if and only if $\det(\nabla^2 v) = 0$. Moreover, if v satisfies the condition $|U^{-1}e_3| \|\nabla v A^{-1}\|_{L^\infty} < \frac{1}{2}$, then ϕ can be chosen such that $u := U(\phi - x', 0)^T$ satisfies the following estimates:

$$\|\nabla^2 u\|_{L^2} \le C \|\nabla v\|_{L^\infty} \|\nabla^2 v\|_{L^2}\,, \tag{46}$$

$$\|u\|_{W^{2,2}} \le C \|\nabla v\|_{L^\infty} \|\nabla^2 v\|_{L^2} + C \|\nabla v\|^2_{L^2}\,. \tag{47}$$

Finally, if $v \in W^{2,\infty}(S)$, then $u \in W^{2,\infty}(S;\mathbb{R}^2)$ and the following inequality holds:

$$\|u\|_{W^{2,\infty}} \le C(\|\nabla^2 v\|_{L^\infty} \|\nabla v\|_{L^\infty} + \|\nabla v\|^2_{L^\infty})\,. \tag{48}$$

Proof. Observe that y of the form (45) satisfies $\nabla y^T \nabla y = A^2$ if and only if

$$\nabla \phi^T A^2 \nabla \phi + |U^{-1}e_3|^2 \nabla v \otimes \nabla v = A^2\,.$$

Defining $\phi' := A\phi$ and $v' = |U^{-1}e_3| v$ the previous equation is equivalent to solve

$$(\nabla \phi' A^{-1})^T (\nabla \phi' A^{-1}) + (\nabla v' A^{-1})^T \otimes (\nabla v' A^{-1})^T = \mathrm{Id}_2$$

for the unknown ϕ'. Define now $\tilde{v}' \in W^{2,2}(AS;\mathbb{R}^3)$ given by $\tilde{v}'(Ax') = v'(x')$. Solving the above equation for ϕ' is equivalent to solve

$$(\nabla \tilde{\phi}')^T \nabla \tilde{\phi}' + \nabla \tilde{v}' \otimes \nabla \tilde{v}' = \mathrm{Id}_2\,,$$

for $\tilde{\phi}' \in W^{1,2}(AS;\mathbb{R}^2)$. Thus, we can conclude by applying [15, Theorem 7]. Note that the transformations $v \rightsquigarrow v' \rightsquigarrow \tilde{v}'$ preserve the property of having null determinant of the Hessian, namely

$$\det(\nabla^2 v) = 0 \iff \det(\nabla^2 v') = 0 \iff \det(\nabla^2 \tilde{v}') = 0\,.$$

Moreover, they preserve the boundedness of the gradient in the L^∞ norm. Finally, define $\tilde{u}' = \tilde{\phi}' - x'$. It is easy to show that

$$\begin{aligned}
\|\nabla u\|_{L^2} &= C\|\nabla \tilde{u}'\|_{L^2}\,,\\
\|u\|_{W^{1,2}} &= C\|\tilde{u}'\|_{W^{1,2}}\,,\\
\|\nabla v\|_{L^\infty} &= C\|\nabla \tilde{v}'\|_{L^\infty}\,,\\
\|\nabla^2 v\|_{L^2} &= C\|\tilde{v}'\|_{L^2}\,.
\end{aligned}$$

Estimates (46)-(47) then follow from [15, Equations (180)–(181)]. We are left to prove (48). Note that it is sufficient to do the proof for $\tilde{u}'$, so we can suppose $U = \mathrm{Id}$. Then, the result follows from [24, Lemma 4.3]. ∎

Lemma A.3. *Let $U \in \mathbb{R}^{3\times 3}$ be a symmetric and positive definite matrix and let $A = \sqrt{(U^2)'}$. Suppose that S is simply connected and satisfies (2). Let $v \in W^{2,2}(S)$ be such that*

$$|U^{-1}e_3|\|\nabla v A^{-1}\|_{L^\infty} < 1$$

and $\det(\nabla^2 v) = 0$. Then there is a sequence $(v_n) \subset C^\infty(\bar{S})$ such that $\det(\nabla^2 v_n) = 0$ for every $n \in \mathbb{N}$ and $v_n \to v$ in $W^{2,2}(S)$.

Proof. Applying Lemma A.2, we construct ϕ such that

$$y(x') = vU^{-1}e_3 + U\begin{pmatrix}\phi\\ 0\end{pmatrix}$$

satisfies $\nabla y^T \nabla y = (U^2)'$. Using the same notation as in the proof of Lemma A.2, we note that ϕ is defined in such a way that $\tilde{\phi}' := A\phi(Ax)$ gives an isometric embedding of $AS \subset \mathbb{R}^2$ for the Euclidean metric, that is, $\nabla\tilde{y}^T\nabla\tilde{y} = \mathrm{Id}_2$, where $\tilde{y} = (\tilde{\phi}', \tilde{v}')^T$. Since (2) holds true also for the set AS, by [17, Theorem 1], there is a sequence of maps $(\tilde{y}_n) \subset W^{2,2}_{\mathrm{iso}}(AS;\mathbb{R}^3) \cap C^\infty(\overline{AS};\mathbb{R}^3)$ such that $\tilde{y}_n \to \tilde{y}$ in $W^{2,2}(AS;\mathbb{R}^3)$, where

$$W^{2,2}_{\mathrm{iso}}(AS;\mathbb{R}^3) = \{y \in W^{2,2}(AS;\mathbb{R}^3)\colon \nabla y^T\nabla y = \mathrm{Id}_2\}\,.$$

Let $\tilde{v}_n = \tilde{y}_n \cdot e_3$. By [15, Theorem 7], we deduce that $\det(\nabla^2\tilde{v}_n) = 0$ for every $n \in \mathbb{N}$. Clearly $\tilde{v}_n \to \tilde{v}$ in $W^{2,2}(AS)$, so $v_n(x) = |U^{-1}e_3|\tilde{v}_n(Ax) \to v$ in $W^{2,2}(S)$. Lastly, $\det(\nabla^2 v_n) = \det(|U^{-1}e_3|\nabla^2\tilde{v}_n A^{-2}) = 0$. ∎

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